

Predictive and Generative AI

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone Expo

Without opening the old lesson, explain one key idea from Capstone Expo. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compute the distribution for Generative and predictive objectives. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

A system retrieves an approved Moon note, uses a trained classifier to predict reading difficulty, then writes a poem. Identify each mechanism and the evidence needed to check its output.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

6. Solve the given case

A tool labels a question biology, writes a new question and computes 7 times 8.

Identify each operation.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 1 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Identify the objective

Classify these components by target: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

My reasoning and evidence

13. W2 Compute the distribution

For The marker is, calculate the three empirical probabilities, their sum and their meaning.

My reasoning and evidence

14. W3 Follow the prefix

After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

My reasoning and evidence

15. W4 Analyse a new corpus independently

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high. State what the most frequent continuation does and does not establish.

My reasoning and evidence

16. W5 Design a useful evaluation

A system reports current room status using retrieved measurements and generated sentences. Name checks for evidence, response quality and permission.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Predictive and Generative AI

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

A fitted classifier labels a question science; its training used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: "Water evaporates into vapour and always disappears forever."

AI ACADEMY / AI-3/1

Predictive and Generative AI

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predict category

The classifier predicts science from the new question.



2. Retrieve note

N supplies a specific statement about evaporation.



3. Generate draft

The draft adds "always disappears forever" beyond N.



4. Verify claims

Keep the supported change-of-state claim; remove or review the unsupported addition.

CHANGE ONE THING

The classifier instead predicts history while N and the draft stay unchanged.

Explain the changed result using the matching lesson.

Identify the first unsupported claim and explain what source would be needed before asserting it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Verified Study Buddy, separate routing predictions, source retrieval and generated explanations in its evidence record.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Verified Study Buddy, separate routing predictions, source retrieval and generated explanations in its evidence record.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.
